MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 10** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If the perimeter of a semicircular region is 72 cm, then its diameter is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 14 cm (B) 28 cm (C) 36 cm (D) 42 cm

Q2. In a right triangle ABC with legs a and b, semicircles are drawn on a, b, and the hypotenuse c. The sum of the areas of the two crescents (lunes) outside the hypotenuse semicircle equals: [1]

(A) Area($\triangle ABC$) (B) $2\times$Area($\triangle ABC$) (C) $\frac{1}{2}$Area($\triangle ABC$) (D) $\pi\times$Area($\triangle ABC$)

Q3. A silver wire brooch in the form of a circle of diameter 35 mm has 5 diameters. The total length of wire used is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 285 mm (B) 110 mm (C) 175 mm (D) 385 mm

Q4. The area of a circle is 154 cm^2. The side of the largest square that can be inscribed in this circle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) $7\sqrt{2}$ cm (B) 14 cm (C) 7 cm (D) $14\sqrt{2}$ cm

Q5. The angle swept by the hour hand of a clock between 2:00 pm and 2:30 pm is: [1]

(A) 15° (B) 30° (C) 180° (D) 7.5°

Q6. **Assertion (A):** The area of a sector of a circle of radius 6 cm with central angle 60° is $\frac{132}{7}$ cm^2.
Reason (R): The area of a sector of central angle θ (in degrees) and radius r is given by Area $= \frac{\theta}{360^\circ} \times \pi r^2$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. A steel wire bent in the form of a square encloses an area of 121 cm^2. If the same wire is bent into a circle, find the area enclosed by the circle. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

Q8. Find the area of the incircle of an equilateral triangle of side $14\sqrt{3}$ cm. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. A circular round table of radius 10 cm has a central square inscribed in it. The four circular segments outside the square are decorated with carvings. Find the total area of the four carved segments. (Use $\pi = 3.14$) [3]

Q10. The inner and outer diameters of a circular running track are 70 m and 84 m respectively. Find the cost of leveling the track at the rate of 5 per m^2. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

Q11. In a right triangle ABC right-angled at B, $AB = 6$ cm and $BC = 8$ cm. Semicircles are described on AB, BC, and the hypotenuse AC as diameters.

(a) Find the hypotenuse AC and the area of $\triangle ABC$.

(b) Find the sum of the areas of the semicircles on AB and BC.

(c) Prove that the area of the two shaded crescents (lunes) formed outside the semicircle on AC is exactly equal to the area of $\triangle ABC$.

[5]

Q12. An equilateral triangle ABC has side 12 cm. A circle is inscribed in it and another circle circumscribes it.

(a) Find the radius and area of the inscribed circle.

(b) Find the radius and area of the circumscribed circle.

(c) Find the area of the circular ring between the circumscribed and inscribed circles.

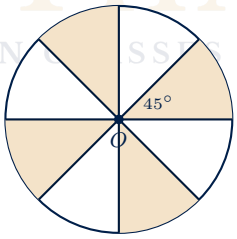
(d) Prove that the area of the circumscribed circle is exactly four times the area of the inscribed circle. (Use $\pi = 3.14$)

[5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Eight-Ribbed Canopy Dome Umbrella Structural Geometry

A canopy umbrella manufacturer constructs premium umbrellas featuring 8 equally spaced flexible ribs. In its fully open position, the umbrella fabric can be approximated as a flat planar circular disc of radius $r = 45$ cm.



(i) Determine the angle formed at the apex centre between any two consecutive ribs. [1]

(ii) Calculate the length of the curved rim arc connecting the tips of two consecutive ribs. [1]

(iii) Calculate the fabric surface area contained between two consecutive ribs, OR calculate the total circular fabric area of the open canopy. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

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